Characterizing Datasets: Hardness, Learning, and More

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Abstract

We develop elementary Fourier analysis *over a finite dataset* $\mathcal{D} \subset \mathcal{T}^n$: coefficients are taken with respect to the empirical measure on $\mathcal{D}$ while the characters remain those of the ambient product space, so that every quantity is estimable from samples. Parseval fails over a dataset — by exactly the density factor $C_{\mathcal{D}} = |\mathcal{T}|^n \ / \ |\mathcal{D}|$ — and we give the correct replacements: a mass identity, a closeness identity, and a low-degree learning theorem in which the normalization constants cancel through a scaled reconstruction. As the highlight, we adapt the Goldreich-Levin algorithm to find all characters heavily correlated with a function *over the dataset*. Sample access alone is provably blind: over a collision-free dataset, every bucket weight in the classical tree search is identical. Query access to a model together with the dataset's heavy bias spectrum suffices, via a convolution identity expressing the on-dataset spectrum as the global spectrum convolved with the spectrum of the dataset itself. Corollaries include a Kushilevitz-Mansour-style guarantee for decision trees over datasets.

1 Fourier Analysis on a Dataset

The classical theory of Boolean functions — and, more generally, Fourier analysis over product probability spaces — is one of the most versatile toolkits in theoretical computer science. Its basic premise, however, is that expectations are taken over a *product measure*: every coordinate is resampled independently. Real data is not like this. A corpus of token sequences, a folder of images, a table of feature vectors — each is a finite set $\mathcal{D} \subset \mathcal{T}^n$ inside an ambient product space, and the functions we care about (a trained model, a labeling rule) are only meaningfully evaluated *on the data*.

This paper develops the elementary theory that results from one stubborn design decision: **keep the characters of the ambient product space, but take all expectations over the dataset.** The resulting "dataset Fourier coefficients" are easy to estimate from samples and retain much of the classical dictionary — but not all of it, and the failures are precise, quantifiable, and (as we will see in the Goldreich-Levin section) algorithmically meaningful. Throughout, think of f as either a trained model, an ideal labeling function on the dataset, or some other efficiently computable function.

1.1 Orthonormal Bases and Parseval

We first fix the ambient framework, which is standard [31].

Definition 1.1 *(Probability Space with Orthonormal Basis)*: Let Ω be a measurable space with probability measure μ, and let I be an index set. An *orthonormal basis* for $L^2(\Omega, \mu)$ is a collection $\{\varphi_\alpha\}_{\alpha \in I}$ of functions $\varphi_\alpha : \Omega \to \mathbb{R}$ satisfying:
1. **Orthonormality:** $\langle \varphi_\alpha, \varphi_\beta \rangle_\mu = \mathbb{E}_{x \sim \mu}[\varphi_\alpha(x)\varphi_\beta(x)] = \delta_{\alpha\beta}$;
2. **Completeness:** every $f \in L^2(\Omega, \mu)$ can be written as $f = \sum_{\alpha \in I} \hat{f}(\alpha)\varphi_\alpha$.

We always take $\varphi_0 = 1$. For product spaces Ω^n with product measure $\mu^{\otimes n}$, the products $\varphi_\alpha = \prod_i \varphi_{\alpha_i}$ (over multi-indices α) form an orthonormal basis for $L^2(\Omega^n, \mu^{\otimes n})$.

Definition 1.2 *(Fourier Coefficients)*: For $f \in L^2(\Omega^n, \mu)$, the *Fourier coefficient* at α is

$$\hat{f}(\alpha) = \langle f, \varphi_\alpha \rangle_\mu = \mathbb{E}_{x \sim \mu}[f(x)\varphi_\alpha(x)],$$

so that $f = \sum_\alpha \hat{f}(\alpha)\varphi_\alpha$ (in L^2).

Proposition 1.1 *(Parseval)*: For any $f, g \in L^2(\Omega^n, \mu)$,

$$\langle f, g \rangle_\mu = \sum_\alpha \hat{f}(\alpha)\hat{g}(\alpha), \quad \text{in particular} \quad \mathbb{E}_{x \sim \mu}[f(x)^2] = \sum_\alpha \hat{f}(\alpha)^2.$$

Proof: Expand both functions in the basis and use orthonormality; the interchange of sum and integral is justified by L^2 convergence. ∎

The two running examples to keep in mind:

Example: *(Haar / parity basis.)* $\Omega = \{-1, 1\}$ with the uniform measure. For $S \subseteq [n]$ the *parity functions* $\chi_S(x) = \prod_{i \in S} x_i$ form an orthonormal basis of $L^2(\{-1,1\}^n)$: for $S \neq T$ pick $i \in S\Delta T$ and note $\mathbb{E}[\chi_{S\Delta T}] = \mathbb{E}[x_i] \cdot \mathbb{E}\left[\chi_{S\Delta T \setminus \{i\}}\right] = 0$.

Example: *(Hermite basis.)* $\Omega = \mathbb{R}$ with the standard Gaussian measure γ. The orthonormalized *probabilist's Hermite polynomials* $h_0 = 1$, $h_1(x) = x$, $h_2(x) = (x^2 - 1)/\sqrt{2}, \ldots$ form an orthonormal basis; their products $H_\alpha(x) = \prod_i h_{\alpha_i}(x_i)$, $\alpha \in \mathbb{N}^n$, handle Gaussian space $L^2(\mathbb{R}^n, \gamma_n)$, with $\deg H_\alpha = |\alpha| = \sum_i \alpha_i$. (The Ornstein-Uhlenbeck noise operator and the invariance principle port over as well; we will not need them here.)

1.2 Coordinate Averaging and Sensitivity

For product spaces there is a standard way to measure how much f depends on a coordinate.

Definition 1.3 *(Coordinate Averaging and Sensitivity)*: For $i \in [n]$, the *averaging operator* is $E^i[f](x) = \mathbb{E}_{a \sim \mu_i}[f(x^{i \leftarrow a})]$, where $x^{i \leftarrow a}$ replaces the i-th coordinate of x by a. The *sensitivity of coordinate* i (often called *influence*) and the *total sensitivity* are

$$\mathrm{Sens}_i[f] = \mathbb{E}_{x \sim \mu}\left[\left(f(x) - E^i[f](x)\right)^2\right], \qquad \boldsymbol{S}[f] = \sum_{i=1}^n \mathrm{Sens}_i[f].$$

Proposition 1.2: E^i kills exactly the basis directions that use coordinate i: $E^i[\varphi_\alpha] = \varphi_\alpha$ if $\alpha_i = 0$ and $E^i[\varphi_\alpha] = 0$ otherwise. Consequently $\mathrm{Sens}_i[f] = \sum_{\alpha : \alpha_i \neq 0} \hat{f}(\alpha)^2$ and $\boldsymbol{S}[f] = \sum_\alpha \#\alpha \cdot \hat{f}(\alpha)^2$, where $\#\alpha = |\{i : \alpha_i \neq 0\}|$.

Proof: The first claim is orthonormality in the i-th factor ($\mathbb{E}_{a \sim \mu_i}\left[\varphi_{\alpha_i}(a)\right] = \left\langle \varphi_{\alpha_i}, \varphi_0 \right\rangle = \delta_{\alpha_i, 0}$); the rest is Parseval applied to $f - E^i f = \sum_{\alpha : \alpha_i \neq 0} \hat{f}(\alpha)\varphi_\alpha$. ∎

This classical quantity is the spectral form of the total-effect (Sobol) index of variance-based global sensitivity analysis [26, 35]; estimated over data, it underlies variance-based feature attribution [12] — a connection the dataset version below makes literal.

1.3 Datasets: the Basic Dictionary

Now for the object of study. Fix a finite alphabet $\mathcal{T}$ and — for this subsection through the end of the paper — let μ be the **uniform** measure on $\mathcal{T}$, with $\{\varphi_\alpha\}$ any orthonormal basis of $L^2(\mathcal{T}^n)$ containing $\varphi_0 = 1$. (We remark below on what survives for non-uniform μ.)

Definition 1.4 *(Dataset; Dataset Fourier Coefficient)*: A *dataset* is a finite nonempty set $\mathcal{D} \subset \mathcal{T}^n$; we write $x \sim \mathcal{D}$ for a uniformly random element of $\mathcal{D}$ and

$$\hat{f}_{\mathcal{D}}(\alpha) = \mathbb{E}_{x\sim\mathcal{D}}[f(x)\varphi_\alpha(x)] = \frac{1}{|\mathcal{D}|}\sum_{x\in\mathcal{D}} f(x)\varphi_\alpha(x)$$

for the *dataset Fourier coefficient* of $f : \mathcal{D} \to \mathbb{R}$ at α. The *density constant* is $C_{\mathcal{D}} = |\mathcal{T}|^n \;/\; |\mathcal{D}|$.

Dataset coefficients are exactly the quantities estimable from labeled samples $(x, f(x))$, $x \sim \mathcal{D}$ — this is the point of the definition. The price: $\{\varphi_\alpha\}$ is orthonormal in $L^2(\mathcal{T}^n, \mu)$, *not* in $L^2(\mathcal{D})$. Two characters may even coincide as functions on $\mathcal{D}$ (we call this *aliasing*; the subcube example in the next section makes it vivid). So none of the Parseval-based dictionary may be borrowed blindly. The correct dictionary is routed through one object:

Definition 1.5 *(The Lift)*: For $f : \mathcal{D} \to \mathbb{R}$, the *lift* $\mathcal{D} \circ f : \mathcal{T}^n \to \mathbb{R}$ is

$$(\mathcal{D} \circ f)(x) = \begin{cases} f(x) & \text{if } x \in \mathcal{D} \\ 0 & \text{otherwise.} \end{cases}$$

Lemma 1.1 *(Lifting Lemma)*: $\widehat{(\mathcal{D} \circ f)}(\alpha) = C_{\mathcal{D}}^{-1} \hat{f}_{\mathcal{D}}(\alpha)$ for every α.

Proof: $\mathbb{E}_{x\sim\mu}[(\mathcal{D} \circ f)(x)\varphi_\alpha(x)] = \frac{1}{|\mathcal{T}|^n}\sum_{x\in\mathcal{D}} f(x)\varphi_\alpha(x) = \frac{|\mathcal{D}|}{|\mathcal{T}|^n} \cdot \hat{f}_{\mathcal{D}}(\alpha)$. ■

Everything true about datasets in this paper is the Lifting Lemma composed with an honest fact about $L^2(\mu)$. Three instances follow. First, inversion — the dataset coefficients determine f on $\mathcal{D}$:

Proposition 1.3 *(Inversion)*: For every $x \in \mathcal{D}$: $\quad f(x) = C_{\mathcal{D}}^{-1} \sum_\alpha \hat{f}_{\mathcal{D}}(\alpha)\varphi_\alpha(x)$.

Proof: Expand the lift in $L^2(\mu)$ and evaluate at $x \in \mathcal{D}$, where $(\mathcal{D} \circ f)(x) = f(x)$; apply Lemma 1.1. ■

Second, the replacement for Parseval. There is no Parseval identity over $\mathcal{D}$; what is true is off by exactly the density constant.

Theorem 1.1 (Mass Identity — "Parseval over a dataset"): For any dataset $\mathcal{D}$ (distinct points) and any $f : \mathcal{D} \to \mathbb{R}$,

$$\sum_\alpha \hat{f}_{\mathcal{D}}(\alpha)^2 = C_{\mathcal{D}} \cdot \mathbb{E}_{x\sim\mathcal{D}}[f(x)^2].$$

Proof: Since $\{\varphi_\alpha\}$ is a complete orthonormal system for the uniform measure on the finite set $\mathcal{T}^n$, it satisfies the *reproducing kernel identity*

$$\sum_{\alpha} \varphi_{\alpha}(x)\varphi_{\alpha}(x') = |\mathcal{T}|^n \cdot \mathbb{1}[x = x']$$

(the matrix $M_{\alpha,x} = \varphi_{\alpha}(x)/\sqrt{|\mathcal{T}|^n}$ is square with orthonormal rows, hence orthonormal columns[1]). Now expand and swap sums:

$$\sum_{\alpha} \hat{f}_{\mathcal{D}}(\alpha)^2 = \frac{1}{|\mathcal{D}|^2} \sum_{x,x' \in \mathcal{D}} f(x)f(x') \sum_{\alpha} \varphi_{\alpha}(x)\varphi_{\alpha}(x') = \frac{|\mathcal{T}|^n}{|\mathcal{D}|^2} \sum_{x \in \mathcal{D}} f(x)^2 = C_{\mathcal{D}} \cdot \mathbb{E}_{\mathcal{D}}[f^2].$$

■

For a non-uniform finite base measure with full support the same proof gives the weighted form $\sum_{\alpha} \hat{f}_{\mathcal{D}}(\alpha)^2 = \frac{1}{|\mathcal{D}|^2} \sum_{x \in \mathcal{D}} f(x)^2/\mu(x)$; for continuous Ω (e.g., Gaussian space) the left side diverges — the empirical measure of a finite dataset has no $L^2(\mu)$ density. This is why we fix a finite alphabet.

Since $C_{\mathcal{D}}$ is enormous for small datasets, the total dataset Fourier mass is *not* $\mathbb{E}_{\mathcal{D}}[f^2]$; a function of unit norm on the dataset carries $C_{\mathcal{D}}$ worth of squared coefficients, smeared across aliases. The constant, however, is not a defect to be endured — it is a normalization to be absorbed. In the language of frame theory, Theorem 1.1 says precisely that the restricted characters $\{\varphi_{\alpha}|_{\mathcal{D}}\}$ form a *tight frame* for $L^2(\mathcal{D})$ with frame bound $C_{\mathcal{D}}$; dividing each frame vector by $\sqrt{C_{\mathcal{D}}}$ yields a *Parseval frame*, for which every identity we care about holds over the dataset with no constants at all.

Definition 1.6 *(Normalized Dataset Coefficients)*: The *normalized characters* and *normalized dataset coefficients* are

$$\breve{\varphi}_{\alpha} = C_{\mathcal{D}}^{-1/2} \varphi_{\alpha}, \qquad \breve{f}_{\mathcal{D}}(\alpha) = \langle f, \breve{\varphi}_{\alpha} \rangle_{\mathcal{D}} = C_{\mathcal{D}}^{-1/2} \hat{f}_{\mathcal{D}}(\alpha).$$

Theorem 1.2 (Parseval and Closeness over the Dataset): For all $f, g : \mathcal{D} \to \mathbb{R}$:
1. (Parseval) $\sum_{\alpha} \breve{f}_{\mathcal{D}}(\alpha)^2 = \mathbb{E}_{x \sim \mathcal{D}}[f(x)^2]$;
2. (Closeness) $\mathbb{E}_{x \sim \mathcal{D}}[(f(x) - g(x))^2] = \sum_{\alpha} \left(\breve{f}_{\mathcal{D}}(\alpha) - \breve{g}_{\mathcal{D}}(\alpha)\right)^2$;
3. (Reconstruction) $f(x) = \sum_{\alpha} \breve{f}_{\mathcal{D}}(\alpha) \breve{\varphi}_{\alpha}(x)$ for every $x \in \mathcal{D}$.

Proof: (1) is Theorem 1.1 divided by $C_{\mathcal{D}}$; (2) is (1) applied to $f - g$, using linearity of $\mathbb{E}_{\mathcal{D}}$; (3) is Proposition 1.3, with the two factors of $C_{\mathcal{D}}^{-1/2}$ recombining into $C_{\mathcal{D}}^{-1}$. ■

Everything on both sides of Theorem 1.2 is over the dataset — no expectation over $\mathcal{T}^n$ appears — and there are no stray constants. The normalization is doing exactly one thing: bookkeeping the overcompleteness. ($L^2(\mathcal{D})$ has dimension $|\mathcal{D}|$, yet the frame has $|\mathcal{T}|^n$ vectors, whose squared norms over $\mathcal{D}$ *average* exactly $C_{\mathcal{D}}^{-1}$ by the kernel diagonal $\sum_{\alpha} \varphi_{\alpha}(x)^2 = |\mathcal{T}|^n$; aliasing is this overcompleteness made visible.) The raw coefficient $\hat{f}_{\mathcal{D}}(\alpha) = \mathbb{E}_{\mathcal{D}}[f\varphi_{\alpha}]$ remains the *estimable correlation* — the quantity a sampling algorithm sees, and the natural unit for heavy-coefficient search in the Goldreich-Levin section — while $\breve{f}_{\mathcal{D}}(\alpha)$ is the natural unit for energy and closeness; translating between them costs exactly one factor of $\sqrt{C_{\mathcal{D}}}$.

Definition 1.7 *($\epsilon_{\mathcal{D}}$-closeness; Normalized Weights)*: We say g is *$\epsilon_{\mathcal{D}}$-close* to f if $\mathbb{E}_{x \sim \mathcal{D}}[(f(x) - g(x))^2] \le \epsilon$. The *normalized level weights* are

[1]Squareness: since the uniform measure has full support, orthonormality forces $|I| \le |\mathcal{T}|^n$, and completeness forces $|I| = |\mathcal{T}|^n$.

$$\overline{W}^k_{\mathcal{D}}[f] = \sum_{\#\alpha=k} \breve{f}_{\mathcal{D}}(\alpha)^2, \quad \text{so that} \quad \sum_{k\geq 0} \overline{W}^k_{\mathcal{D}}[f] = \mathbb{E}_{\mathcal{D}}[f^2]$$

by Theorem 1.2.

1.4 Averaging and Sensitivity on a Dataset

The third instance of the lift-then-use-$L^2(\mu)$ recipe concerns coordinate averaging. On a dataset, resampling a coordinate may leave the data; the natural operator zeros out such settings — and this choice makes it agree exactly with the global operator applied to the lift.

Definition 1.8 *(Dataset Coordinate Averaging; Dataset Sensitivity)*: For $x \in \mathcal{D}$,

$$E^i_{\mathcal{D}}[f](x) = \mathbb{E}_{a\sim\mu_i}[\mathcal{D}(x^{i\leftarrow a}) \cdot f(x^{i\leftarrow a})], \qquad \mathrm{Sens}^i_{\mathcal{D}}[f] = \mathbb{E}_{x\sim\mathcal{D}}\Big[\big(f(x) - E^i_{\mathcal{D}}[f](x)\big)^2\Big],$$

where $\mathcal{D}(\cdot)$ denotes the indicator of the dataset. Note $E^i_{\mathcal{D}}[f] = E^i[\mathcal{D} \circ f]$ restricted to $\mathcal{D}$, directly from the definitions.

Proposition 1.4 *(Dataset Sensitivity Bound)*:

$$\mathrm{Sens}^i_{\mathcal{D}}[f] \leq C_{\mathcal{D}} \cdot \mathbb{E}_{x\sim\mu}\Big[\big((\mathcal{D} \circ f) - E^i[\mathcal{D} \circ f]\big)^2\Big] = \sum_{\alpha:\alpha_i\neq 0} \breve{f}_{\mathcal{D}}(\alpha)^2,$$

and the inequality is an equality when $(\mathcal{D} \circ f) - E^i[\mathcal{D} \circ f]$ is supported on $\mathcal{D}$. Consequently $\boldsymbol{S}_{\mathcal{D}}[f] := \sum_i \mathrm{Sens}^i_{\mathcal{D}}[f] \leq \sum_{k\geq 1} k \cdot \overline{W}^k_{\mathcal{D}}[f]$.

Proof: Write $q = (\mathcal{D} \circ f) - E^i[\mathcal{D} \circ f]$, so that $\mathrm{Sens}^i_{\mathcal{D}}[f] = \mathbb{E}_{\mathcal{D}}[q^2] = C_{\mathcal{D}} \cdot \mathbb{E}_\mu[\mathcal{D} \cdot q^2] \leq C_{\mathcal{D}} \cdot \mathbb{E}_\mu[q^2]$. Then apply Proposition 1.2 to the lift and Lemma 1.1; summing over i counts each α once per nonzero coordinate. ∎

The inequality can be strict, and the naive analogue of Proposition 1.2 with dataset coefficients (no $C_{\mathcal{D}}^{-1}$) is simply false. A useful example to remember: $\mathcal{D} = \{x : x_1 = 1\} \subset \{-1, 1\}^n$ and $f = x_2$. Then $\mathrm{Sens}^2_{\mathcal{D}}[f] = 1$, while $\chi_{\{2\}}$ and $\chi_{\{1,2\}}$ alias on $\mathcal{D}$, so $\sum_{\alpha:\alpha_2\neq 0} \hat{f}_{\mathcal{D}}(\alpha)^2 = 1 + 1 = 2$; in normalized units the bound of Proposition 1.4 reads $\sum_{\alpha:\alpha_2\neq 0} \breve{f}_{\mathcal{D}}(\alpha)^2 = 2/C_{\mathcal{D}} = 1$ (here $C_{\mathcal{D}} = 2$) — tight.

1.5 Learning Low-Degree Functions from Dataset Samples

As a first payoff, the classical Low-Degree Algorithm [31] survives on datasets, *provided* the reconstruction is the truncated *frame* expansion $\sum_{\#\alpha\leq d} \breve{f}_{\mathcal{D}}(\alpha)\breve{\varphi}_\alpha$ — equivalently, the raw plug-in reconstruction scaled by $C_{\mathcal{D}}^{-1}$. The scaling matters: in the half-cube example above, the unscaled plug-in reconstruction of $f = x_2$ at degree 2 returns $\chi_2 + \chi_{12} = 2x_2$ on $\mathcal{D}$ — error 1 with exact coefficients and zero tail — because each character is counted once per alias. The frame reconstruction returns $(x_2 + x_1x_2)/2 = x_2$ on $\mathcal{D}$, exactly — by part (3) of Theorem 1.2 together with the fact that all coefficients outside $\{\chi_2, \chi_{12}\}$ vanish there.

Two standing conventions for the statement. The one-coordinate basis is assumed B-*bounded* — $\|\varphi_j\|_\infty \leq B$ for all j, so that $\|\varphi_\alpha\|_\infty \leq B^d$ whenever $\#\alpha \leq d$; for the parity characters, and for any group-character basis, $B = 1$. And we write $N_d = |\{\alpha : \#\alpha \leq d\}| = \sum_{k\leq d}\binom{n}{k}(|\mathcal{T}| - 1)^k \leq (1 + n(|\mathcal{T}| - 1))^d$ for the number of low-degree indices.

Theorem 1.3 (Learning Low-Degree Functions over a Dataset): Let $f : \mathcal{D} \to [-1, 1]$ and suppose the normalized high-degree weight satisfies

$$\sum_{k>d} \overline{W}^k_{\mathcal{D}}[f] \le \frac{\epsilon}{4}$$

(e.g., $d \ge 4\overline{S}/\epsilon$ suffices when $\sum_k k \cdot \overline{W}^k_{\mathcal{D}}[f] \le \overline{S}$). Given $|\mathcal{D}|$ and $m = O\Big(B^{2d} \frac{N_d}{\epsilon} \cdot \log(N_d/\delta)\Big)$ samples from $\mathcal{D}$, one can output h that is $\epsilon_{\mathcal{D}}$-close to f with probability at least $1 - \delta$.

Proof: **Algorithm.** For each $\#\alpha \le d$, estimate $\hat{f}_{\mathcal{D}}(\alpha)$ by $\tilde{f}(\alpha) = \frac{1}{m} \sum_{j=1}^{m} f\big(x^{(j)}\big)\varphi_\alpha\big(x^{(j)}\big)$; output the *scaled* reconstruction $h = C_{\mathcal{D}}^{-1} \sum_{\#\alpha \le d} \tilde{f}(\alpha)\varphi_\alpha$ (this scaling is where knowledge of $|\mathcal{D}|$ enters).

Analysis. Write $g^{\uparrow} = \mathcal{D} \circ f$, so $\hat{g^{\uparrow}}(\alpha) = C_{\mathcal{D}}^{-1} \hat{f}_{\mathcal{D}}(\alpha)$ (Lemma 1.1) and $f = g^{\uparrow}$ on $\mathcal{D}$. Dropping the indicator and applying Parseval in $L^2(\mu)$:

$$\begin{aligned} \mathbb{E}_{\mathcal{D}}\big[(f-h)^2\big] = C_{\mathcal{D}} \cdot \mathbb{E}_\mu\Big[\mathcal{D} \cdot \big(g^{\uparrow} - h\big)^2\Big] &\le C_{\mathcal{D}} \cdot \mathbb{E}_\mu\Big[\big(g^{\uparrow} - h\big)^2\Big] \\ &= C_{\mathcal{D}} \sum_{\alpha} \Big(\hat{g^{\uparrow}}(\alpha) - \hat{h}(\alpha)\Big)^2. \end{aligned}$$

Since $\hat{h}(\alpha) = C_{\mathcal{D}}^{-1} \tilde{f}(\alpha)$ for $\#\alpha \le d$ and 0 otherwise, the $C_{\mathcal{D}}$ cancels against the two factors of $C_{\mathcal{D}}^{-1}$:

$$\mathbb{E}_{\mathcal{D}}\big[(f-h)^2\big] \le \underbrace{\sum_{k>d} \overline{W}^k_{\mathcal{D}}[f]}_{\le \epsilon/4 \text{ (hypothesis)}} + \underbrace{C_{\mathcal{D}}^{-1} \sum_{\#\alpha \le d} \Big(\hat{f}_{\mathcal{D}}(\alpha) - \tilde{f}(\alpha)\Big)^2}_{\le \epsilon/4 \text{ w.h.p.}}.$$

For the second term: each $\tilde{f}(\alpha)$ averages m i.i.d. terms bounded by $\|f\|_\infty \|\varphi_\alpha\|_\infty \le B^d$ with mean $\hat{f}_{\mathcal{D}}(\alpha)$; by Hoeffding (range $2B^d$) and a union bound over the N_d low-degree indices, the stated m makes every deviation at most $\frac{1}{2}\sqrt{\epsilon/N_d}$, so the sum is at most $N_d \cdot \epsilon/(4N_d) = \epsilon/4$ even before the (generous) factor $C_{\mathcal{D}}^{-1} \le 1$. ■

Two remarks. First, the hypothesis is stated in normalized weights, the correct analogue of "ϵ of Fourier weight above degree d"; in raw dataset units it reads $\sum_{k>d} \sum_{\#\alpha=k} \hat{f}_{\mathcal{D}}(\alpha)^2 \le C_{\mathcal{D}}\epsilon/4$. Second, for a *generic* sparse dataset the hypothesis is genuinely demanding — it asks the lift $\mathcal{D} \circ f$ to be low-degree concentrated in $L^2(\mu)$ — and the next section shows this is not an artifact of our proof but a structural fact about datasets. Structured datasets (dense ones, or subcube-like ones as in the example above) satisfy it naturally.

1.6 Notation Summary

For reference, the recurring symbols of the paper (the last four families are introduced in the Goldreich-Levin section):

Symbol	**Meaning**
$\hat{f}(\alpha)$	global Fourier coefficient $\mathbb{E}_{x\sim\mu}[f\varphi_\alpha]$
$\hat{f}_{\mathcal{D}}(\alpha)$	dataset coefficient $\mathbb{E}_{x\sim\mathcal{D}}[f\varphi_\alpha]$ — the estimable correlation
$\breve{f}_{\mathcal{D}}(\alpha) = C_{\mathcal{D}}^{-1/2} \hat{f}_{\mathcal{D}}(\alpha)$	normalized (Parseval-frame) coefficient — the unit for energy and closeness

Symbol	Meaning
$C_{\mathcal{D}} = \lvert\mathcal{T}\rvert^n \;/\; \lvert\mathcal{D}\rvert$	density constant; frame bound of $\{\varphi_\alpha\lvert_{\mathcal{D}}\}$
$\mathcal{D} \circ f$	the lift: f on $\mathcal{D}$, 0 elsewhere; $\widehat{(\mathcal{D} \circ f)} = C_{\mathcal{D}}^{-1} \hat{f}_{\mathcal{D}}$
$\overline{W}^k_{\mathcal{D}}[f] = \sum_{\#\alpha=k} \breve{f}_{\mathcal{D}}(\alpha)^2$	normalized level weight; $\sum_k \overline{W}^k = \mathbb{E}_{\mathcal{D}}[f^2]$
b_V, B_θ	bias spectrum of $\mathcal{D}$ (dataset coefficients of 1); its θ-heavy set
$\overline{v}_S(z)$, $\Psi(S\lvert J)$	average with context z fixed; conditional bucket weight $\mathbb{E}_z[\overline{v}_S(z)^2]$
c_k, R_k	context profile (distinct contexts at level k); level mass $\sum_S \Psi$
N_k, N	live-bucket profile: buckets with $\Psi \ge \tau^2/4$ at level k; its maximum

2 Highlight: Goldreich-Levin over a Dataset

We now come to the highlight: adapting the Goldreich-Levin algorithm [18] ([31], Section 3.5) to the dataset setting. The classical algorithm, given query access to f, finds *all* of the characters with which f is noticeably correlated — the engine behind the Kushilevitz-Mansour learning algorithm [29]. Our use case is the on-distribution version of the same question: f is a trained model (or labeling rule), $\mathcal{D}$ is the data, and we want every character that is noticeably correlated with f *over the dataset*, i.e., every heavy dataset coefficient $\hat{f}_{\mathcal{D}}(S)$. Moreover, we insist that the headline algorithm *never leaves the data*: no membership tester for $\mathcal{D}$, and no evaluation of f on synthetic, out-of-distribution inputs. The resource that replaces them is one that real corpora actually offer: *conditional sampling* — retrieving datapoints that share a fixed substring (a context), the way an n-gram index retrieves them from a text corpus.

For clarity of exposition we work over the Boolean cube $\Omega^n = \{-1,1\}^n$ with the parity characters χ_S, writing $\hat{f}_{\mathcal{D}}(S) = \mathbb{E}_{x\sim\mathcal{D}}[f(x)\chi_S(x)]$ as before.[2] The goal, mirroring the classical guarantee:

Definition 2.1 *(Dataset Heavy-Coefficient Search)*: Given $0 < \tau \le 1$ and access to f and $\mathcal{D}$ (in a model to be specified), output a list L of subsets of $[n]$ such that with high probability:

1. (Completeness) $\lvert\hat{f}_{\mathcal{D}}(S)\rvert \ge \tau \Longrightarrow S \in L$;
2. (Soundness) $S \in L \Longrightarrow \lvert\hat{f}_{\mathcal{D}}(S)\rvert \ge \tau/2$.

2.1 Access Models

The choice of access model turns out to be the crux of the problem. The underlying object is a corpus $\mathcal{D} \subset \{-1,1\}^n$ together with f on $\mathcal{D}$; the algorithm reaches it through:

1. **Samples** (SAMP): i.i.d. draws $x \sim \mathcal{D}$ together with the label $f(x)$.
2. **Conditional samples** (CSAMP): given coordinates $\overline{J} \subseteq [n]$ and a context z realized in the data, a draw $x \sim \mathcal{D}$ conditioned on $x_{\overline{J}} = z$, together with $f(x)$. This is the "retrieve strings containing this substring" primitive; it is implemented exactly by an index over the corpus (an n-gram or suffix-array structure), or by holding an explicit copy — or large subsample — of $\mathcal{D}$. It is the *subcube-conditioning* oracle studied in distribution testing and learning [9–

[2] Every identity below that uses only orthonormality and the reproducing kernel $\sum_S \chi_S(x)\chi_S(x') = 2^n \cdot \mathbb{1}[x = x']$ extends verbatim to any finite abelian group with its character basis (with $\chi_S\overline{\chi_T}$ in place of $\chi_S\chi_T$), and to general product orthonormal bases over finite sets. The convolution identity additionally uses the group structure $\chi_S\chi_T = \chi_{S\Delta T}$.

11], specialized here to the empirical support of $\mathcal{D}$ — and put to a different use: recovering heavy Fourier coefficients of a function f, rather than testing a distribution.

That is the entire model. Note what is absent: any membership tester for $\mathcal{D}$, any evaluation of f outside the data, indeed any synthetic input whatsoever — every string the algorithm ever touches is a datapoint. (A companion result, Theorem 2.3, considers the strictly stronger setting where f is additionally a cheaply queryable model; we keep that assumption out of the headline deliberately, to highlight that it is not needed.)

Classical GL is an algorithm of the stronger type — it queries f at arbitrary, algorithm-chosen points — and its guarantee concerns the *global* coefficients $\hat{f}(S)$. For the *dataset* coefficients, we will see that plain samples are provably insufficient (Lemma 2.2); the headline theorem (Theorem 2.2) restores power through CSAMP alone, with complexity governed by how much the dataset's contexts *repeat*, while the companion routes the dataset's structure through its *bias spectrum* (Lemma 2.5) at the price of model queries.

2.2 Recap: the Classical Goldreich-Levin Algorithm

Theorem 2.1 (Goldreich-Levin [18]; [31] Section 3.5): Given query access to $f : \{-1,1\}^n \to \{-1,1\}$ and $0 < \tau \leq 1$, there is a poly$(n, 1/\tau)$-time algorithm that with high probability outputs a list $L = \{U_1, ..., U_\ell\}$ of subsets of $[n]$ such that:
1. $|\hat{f}(U)| \geq \tau \Longrightarrow U \in L$;
2. $U \in L \Longrightarrow |\hat{f}(U)| \geq \tau/2$.

By Parseval, the second guarantee implies $|L| \leq 4/\tau^2$.

The engine is a divide-and-conquer search over the tree of coordinate prefixes. For $S \subseteq J \subseteq [n]$, define the **bucket weight** ([31], Definition 3.39)

$$W^{S \mid J}[f] = \sum_{T \subseteq \overline{J}} \hat{f}(S \cup T)^2,$$

the Fourier weight of f on sets whose restriction to J equals S. The algorithm maintains the buckets of weight at least $\tau^2/2$ — at most $2/\tau^2$ of them by Parseval ($4/\tau^2$ once estimation slack is accounted) — splitting one coordinate at a time until each survivor is a singleton $\{S\}$ with $\hat{f}(S)^2 \geq \tau^2/2$. What makes this feasible is that bucket weights are *estimable*: by the restriction identity ([31], equation (3.5) and Proposition 3.40),

$$W^{S \mid J}[f] = \mathbb{E}_{z \sim \{-1,1\}^{\overline{J}}}\left[\hat{f}_{J \mid z}(S)^2\right] = \mathbb{E}_{z \sim \{-1,1\}^{\overline{J}}}\ \mathbb{E}_{y,y' \sim \{-1,1\}^J}[f(y,z)\chi_S(y) \cdot f(y',z)\chi_S(y')],$$

so an algorithm with query access to f gets accuracy $\pm\epsilon$ with $O(\log(1/\delta)/\epsilon^2)$ queries.

Two remarks for later use. First, the proof only needs $\|f\|_\infty \leq 1$ (for the Chernoff estimates) and $\|f\|_2 \leq 1$ (for the Parseval pruning bound), so Theorem 2.1 holds verbatim for $f : \{-1,1\}^n \to [-1,1]$, with list size $4\|f\|_2^2/\tau^2$. Second, the restriction identity averages the *uniform* restriction z; keep an eye on it — the headline theorem below will replace this uniform average by the dataset's own conditional distribution.

2.3 Obstructions

Why can't we just run the same tree search on the dataset coefficients? Recall from the Mass Identity (Theorem 1.1) that $\sum_S \hat{f}_{\mathcal{D}}(S)^2 = C_{\mathcal{D}} \cdot \mathbb{E}_{\mathcal{D}}[f^2]$: Parseval fails over $\mathcal{D}$ by exactly the density factor. This already kills the pruning bound — the number of τ-heavy dataset coefficients can be as large as $C_{\mathcal{D}}\mathbb{E}_{\mathcal{D}}[f^2]/\tau^2$, i.e., exponential — and the failure is not confined to exotic functions:

Example: Let $K \subseteq [n]$ and let $\mathcal{D} = \{x : x_i = 1 \text{ for all } i \in K\}$ be the subcube fixing K, with $f \equiv 1$. Every χ_S with $S \subseteq K$ is identically 1 on $\mathcal{D}$, so

$$\hat{f}_{\mathcal{D}}(S) = \begin{cases} 1 \text{ if } S \subseteq K \\ 0 \text{ otherwise,} \end{cases}$$

and there are $2^{|K|}$ maximal coefficients: distinct characters are *identical as functions on* $\mathcal{D}$, and "the" heavy set is only defined up to this aliasing.

More damning still: the bucket weights themselves carry *no information* over a generic dataset. The following exact identity is the dataset analogue of the restriction identity.

Lemma 2.1 *(Pair Form of Dataset Bucket Weights)*: Fix $J \subseteq [n]$ with $|J| = k$ and $S \subseteq J$. Then

$$\sum_{U \subseteq \overline{J}} \hat{f}_{\mathcal{D}}(S \cup U)^2 = 2^{n-k} \cdot \mathbb{E}_{x,x' \sim \mathcal{D}}[f(x)f(x')\chi_S(x_J)\chi_S(x'_J) \cdot \mathbb{1}[x_{\overline{J}} = x'_{\overline{J}}]].$$

Proof: Expand each $\hat{f}_{\mathcal{D}}(S \cup U) = \mathbb{E}_{x\sim\mathcal{D}}[f(x)\chi_S(x_J)\chi_U(x_{\overline{J}})]$, square, and swap sums; the inner sum $\sum_{U \subseteq \overline{J}} \chi_U(x_{\overline{J}})\chi_U(x'_{\overline{J}})$ is the reproducing kernel $2^{n-k} \cdot \mathbb{1}[x_{\overline{J}} = x'_{\overline{J}}]$ on $\{-1,1\}^{\overline{J}}$.■

Lemma 2.2 *(Blindness of Dataset Bucket Weights)*: In the setting of Lemma 2.1, suppose no two dataset points collide on $\overline{J}$ (the projection $x \mapsto x_{\overline{J}}$ is injective on $\mathcal{D}$). Then for *every* $S \subseteq J$,

$$\sum_{U \subseteq \overline{J}} \hat{f}_{\mathcal{D}}(S \cup U)^2 = \frac{2^{n-k}}{|\mathcal{D}|} \cdot \mathbb{E}_{x\sim\mathcal{D}}[f(x)^2],$$

independent of S.

Proof: Under injectivity only the diagonal of Lemma 2.1 survives, and $\chi_S(x_J)^2 = 1$. ■

In words: as long as the dataset's projection onto the un-split coordinates is collision-free, every bucket at level k has exactly the same weight — no matter what f is, and no matter where its heavy dataset coefficients lie. For m points in general position, a union bound gives collisions on $\overline{J}$ with probability at most $\binom{m}{2}/2^{n-k}$, so the search tree is information-free for all levels $k \lesssim n - 2\log_2 m$ — the *birthday horizon*. A GL-style search would descend blindly through $2^{n-2\log_2 m}$ buckets before the weights begin to distinguish anything. Together with the subcube example, this rules out a sample-only dataset GL: additional access, or structural repetition in $\mathcal{D}$, is necessary. This is a dataset-specific incarnation of the statistical-query / noisy-parity barrier [6, 7, 28] — correlational (sample-only) access cannot isolate a heavy parity — and of the sample-only insufficiency that interactive proofs for learning circumvent with a query-capable prover [19, 22]; here the missing power will be supplied instead by context repetition (Theorem 2.2) or by the dataset's bias spectrum (Theorem 2.3).[3]

[3] A sanity check for Lemma 2.2: $n = 2$, $\mathcal{D} = \{(1,1), (-1,-1)\}$, $f(x) = x_1$, $J = \{1\}$. Then $\hat{f}_{\mathcal{D}}(\{1\}) = 1$, $\hat{f}_{\mathcal{D}}(\{1,2\}) = 0$, and the bucket weight is $1 = \frac{2^{2-1}}{2} \cdot 1$: the signal coefficient and the aliasing noise conspire to a flat bucket profile. All identities and theorem constants in this section have been verified numerically on random and structured instances.

2.4 Main Theorem: Goldreich-Levin from Context Conditioning

The blindness lemma tells us exactly what to look for: its collision-free hypothesis fails precisely when dataset points *share contexts*. Real corpora share contexts constantly — that is what n-gram statistics are — and CSAMP is the primitive that exposes the sharing. The headline algorithm replaces the classical bucket weight, whose restriction z is averaged over the *uniform* measure, by a conditional statistic whose contexts are drawn from *the dataset itself*.

Definition 2.2 *(Contexts, Fibers, and the Context Profile)*: Fix $J = J_k = [k]$ and $\overline{J} = [n] \setminus J$. The *contexts* at level k are the realized projections $Z_k = \{x_{\overline{J}} : x \in \mathcal{D}\}$; the *fiber* of $z \in Z_k$ is $\mathcal{D}_z = \{x \in \mathcal{D} : x_{\overline{J}} = z\}$. We write $z \sim \mathcal{D}_{\overline{J}}$ for the marginal (i.e., $z = x_{\overline{J}}$ with $x \sim \mathcal{D}$), and define the *context profile* $c_k = |Z_k|$.

Definition 2.3 *(Conditional Bucket Weight)*: For $S \subseteq J$ and $z \in Z_k$, let $\overline{v}_S(z) = \mathbb{E}_{x \sim \mathcal{D}_z}[f(x)\chi_S(x_J)]$ — the "average with the string z fixed." The *conditional bucket weight* is

$$\Psi(S \mid J) = \mathbb{E}_{z \sim \mathcal{D}_{\overline{J}}}[\overline{v}_S(z)^2].$$

Lemma 2.3 *(Properties of the Conditional Bucket Weight)*: For all $S \subseteq J = J_k$:
1. (Tower) $\hat{f}_{\mathcal{D}}(S \cup U) = \mathbb{E}_{z \sim \mathcal{D}_{\overline{J}}}[\chi_U(z)\overline{v}_S(z)]$ for every $U \subseteq \overline{J}$;
2. (Completeness) $\Psi(S \mid J) \geq \hat{f}_{\mathcal{D}}(S \cup U)^2$ for every $U \subseteq \overline{J}$;
3. (Termination) $\Psi(S \mid [n]) = \hat{f}_{\mathcal{D}}(S)^2$;
4. (Monotonicity) $\Psi(S' \mid J_{k+1}) \leq \Psi(S \mid J_k)$ for both children $S' \in \{S, S \cup \{k+1\}\}$;
5. (Level mass) $\sum_{S \subseteq J} \Psi(S \mid J) = 2^k \cdot \mathbb{E}_{z \sim \mathcal{D}_{\overline{J}}}\left[\frac{\mathbb{E}_{x \sim \mathcal{D}_z}[f(x)^2]}{|\mathcal{D}_z|}\right] =: R_k \leq 2^k \cdot \|f\|_\infty^2 \cdot \frac{c_k}{|\mathcal{D}|}$.

Proof: (1) is the tower property of conditional expectation. (2) is Cauchy-Schwarz applied to (1), using $\chi_U(z)^2 = 1$. (3): at $J = [n]$ there is a single trivial context and $\overline{v}_S = \hat{f}_{\mathcal{D}}(S)$. (4): for $S' = S$, coarsening the context from level k to level $k+1$ replaces $\overline{v}_S(z)$ by its conditional average, and Jensen gives $\mathbb{E}[\overline{v}^2]$ non-increasing. For $S' = S \cup \{k+1\}$, write $w = f \cdot \chi_{S'}\left(x_{J_{k+1}}\right)$; the level-$k$ context fixes x_{k+1}, so $\mathbb{E}[w \mid z] = \pm\overline{v}_S(z)$ and $\mathbb{E}_z[\mathbb{E}[w|z]^2] = \Psi(S \mid J_k)$; coarsening to level $k+1$ is again Jensen. (5): points of a fiber share z, hence have *distinct* J-parts, so $\{x_J : x \in \mathcal{D}_z\}$ is a dataset in $\{-1,1\}^J$ and the Mass Identity (Theorem 1.1) on it gives $\sum_{S \subseteq J} \overline{v}_S(z)^2 = \frac{2^k}{|\mathcal{D}_z|}\mathbb{E}_{x \sim \mathcal{D}_z}[f^2]$; average over z. The final bound uses $\mathbb{E}_{z \sim \mathcal{D}_{\overline{J}}}[1/\ |\mathcal{D}_z|] = c_k/\ |\mathcal{D}|$. ∎

Everything a GL search needs is here: heavy descendants keep their ancestors heavy (2), the search terminates at exactly the dataset coefficient (3), pruned buckets stay pruned (4), and the number of live buckets is controlled by the *repetition structure* of the data (5) — R_k is small exactly when contexts repeat ($c_k \ll |\mathcal{D}|$ at levels where 2^k is large). Estimation is the dataset analogue of the classical restriction identity: draw a context *from the data* and two strings sharing it.

Lemma 2.4 *(Estimating Ψ)*:

$$\Psi(S \mid J) = \mathbb{E}_{z \sim \mathcal{D}_{\overline{J}}}\ \mathbb{E}_{x,x' \text{ i.i.d.} \sim \mathcal{D}_z}[f(x)\chi_S(x_J) \cdot f(x')\chi_S(x'_J)].$$

Each experiment costs one SAMP draw (yielding x and its context $z = x_{\overline{J}}$) and one CSAMP draw ($x' \sim \mathcal{D}_z$); the product lies in $[-1,1]$ when $\|f\|_\infty \leq 1$, so $O(\log(1/\delta)/\epsilon^2)$ experiments estimate $\Psi(S|J)$ to accuracy $\pm\epsilon$ with confidence $1-\delta$.

Proof: For fixed z, independence of x, x' gives $\mathbb{E}[f(x)\chi_S(x_J)f(x')\chi_S(x'_J)] = \overline{v}_S(z)^2$; average over z. A SAMP draw x has $x_{\overline{J}} \sim \mathcal{D}_{\overline{J}}$ and, conditioned on its context, $x \sim \mathcal{D}_z$; Hoeffding does the rest. ∎

Definition 2.4 *(Live-Bucket Profile)*: For a threshold τ, the *live-bucket profile* of $(\mathcal{D}, f)$ is

$$N_k = |\{S \subseteq J_k : \Psi(S \mid J_k) \geq \tau^2/4\}|, \qquad N = \max_{0 \leq k \leq n} N_k.$$

By Markov on the level mass (property 5) and monotonicity (property 4), $N_k \leq \min(4R_k/\tau^2,\ 2N_{k-1})$, the latter for $k \geq 1$; the instances after the theorem show how the repetition structure of $\mathcal{D}$ controls N.

Theorem 2.2 (Dataset Goldreich-Levin from Context Conditioning): Given SAMP and CSAMP access to $(\mathcal{D}, f)$ with $\|f\|_\infty \leq 1$, and $0 < \tau \leq 1$, $\delta \in (0,1)$, there is an algorithm that with probability at least $1 - \delta$ outputs a list L satisfying:
1. (Completeness) $|\hat{f}_{\mathcal{D}}(S)| \geq \tau \Longrightarrow S \in L$;
2. (Soundness) $S \in L \Longrightarrow |\hat{f}_{\mathcal{D}}(S)| \geq \tau/2$;
3. (Complexity) on the same event, at most $O(nN \cdot \log(nN/\delta)/\tau^4)$ sampling experiments and $\mathrm{poly}(n, N, 1/\tau, \log(1/\delta))$ time.

The algorithm needs no advance knowledge of N, never evaluates f outside $\mathcal{D}$, and never tests membership: every string it touches is a datapoint.

The algorithm is the classical tree search with Ψ in place of W:
1. Maintain a set of live buckets (S, J_k), starting from $(\emptyset, J_0)$.
2. At level k, split each live bucket on coordinate $k+1$ and estimate $\Psi(S' \mid J_{k+1})$ for both children to accuracy $\tau^2/8$, giving the t-th estimate performed (in any fixed processing order) confidence $1 - \delta/(2t^2)$ via Lemma 2.4; keep the children with estimate at least $3\tau^2/8$. Note the confidence schedule references only the running index t, so the algorithm is well-defined with no advance knowledge of N.
3. Output the surviving singletons at level n.

Proof: Condition on every performed estimate being $\tau^2/8$-accurate. The t-th estimate uses fresh samples, so it fails with probability at most $\delta/(2t^2)$ regardless of which buckets earlier estimates caused to be explored; summing over the (adaptively determined, possibly unbounded) sequence of estimates, the total failure probability is at most $\sum_{t \geq 1} \delta/(2t^2) \leq \delta$.

Completeness. If $|\hat{f}_{\mathcal{D}}(S)| \geq \tau$, then by completeness (2) of Lemma 2.3 every ancestor bucket of S has $\Psi \geq \tau^2$, so its estimate is at least $7\tau^2/8 \geq 3\tau^2/8$ and the ancestor survives at every level; hence $S \in L$.

Soundness. A surviving leaf has estimate at least $3\tau^2/8$, hence $\Psi(S \mid [n]) \geq \tau^2/4$; by termination (3), $\hat{f}_{\mathcal{D}}(S)^2 \geq \tau^2/4$.

Complexity. On the good event, a surviving bucket at level k has true $\Psi \geq \tau^2/4$, so there are at most N_k of them; the level-mass identity (5) gives $N_k \leq 4R_k/\tau^2$ (Markov), and monotonicity (4) gives $N_k \leq 2N_{k-1}$ (children of dead buckets are dead). Hence at most $2nN$ estimates are performed in total, and the t-th costs $O(\log(t/\delta)/\tau^4) = O(\log(nN/\delta)/\tau^4)$ experiments. ∎

(For a deterministic time budget: cap the live buckets per level at $\overline{N}$, abort on overflow, and guess-and-double $\overline{N}$ with confidence budget halved per run; the run with $\overline{N} \geq N$ never aborts on the good event, at the cost of $O(\log N)$ restarts.)

Three instances of the parameter N, spanning the possible regimes (all verified numerically):

- **Dense datasets.** Always $c_k \leq \min(|\mathcal{D}|, 2^{n-k})$, so $R_k \leq \min(2^k, C_{\mathcal{D}})$ and $N \leq 4C_{\mathcal{D}}/\tau^2$: for $C_{\mathcal{D}} = \text{poly}(n)$ the search is polynomial outright — using strictly weaker access than a membership tester.
- **Subcube-structured data.** For the subcube example above (fixing K, $f \equiv 1$): $R_k = 2^{|K \cap J_k|} \leq 2^{|K|}$, matching the true output size $2^{|K|}$ — the search is output-sensitive, even though the density constant $C_{\mathcal{D}} = 2^{|K|}$ is itself exponential in $|K|$.
- **Generic sparse data.** Singleton fibers give $\overline{v}_S(z) = f(x)\chi_S(x_J)$, so $\Psi(S|J) = \mathbb{E}_{\mathcal{D}}[f^2]$ for *every* S — flat — and $c_k = |\mathcal{D}|$, $R_k = 2^k \cdot \mathbb{E}_{\mathcal{D}}[f^2]$: the parameter degrades to the trivial 2^k exactly when Lemma 2.2 bites, and CSAMP degenerates to SAMP. The theorem's power is context repetition, and the blindness lemma says this is not an artifact.

Two further remarks. First, nothing forces the split order $1, 2, ..., n$: any coordinate order works, and the profile R_k — hence the runtime — depends on it. For sequence data the natural choice is to keep *contiguous* contexts (so fibers are "strings sharing an $(n-k)$-gram"), and choosing the order to maximize repetition is a purely combinatorial preprocessing question on $\mathcal{D}$. Second, applying Theorem 2.2 to the constant function $f \equiv 1$ finds all heavy *biases* $b_V = \mathbb{E}_{\mathcal{D}}[\chi_V]$ at threshold θ in time $\text{poly}(n, N(\mathbb{1}), 1/\theta)$ — recovering, on-distribution, exactly the side information that the companion result below consumes.

2.5 Companion: Dataset GL from Model Queries

This subsection — and only this subsection, together with its corollaries below — assumes one additional oracle: **model queries** (QUERY), i.e., f evaluable at arbitrary $x \in \{-1, 1\}^n$, including out-of-distribution points. This is realistic when f is a cheaply queryable model, and it enables a different route: never condition, but instead read the dataset's effect on the spectrum through one structural object.

Definition 2.5 *(Bias Spectrum)*: The **bias spectrum** of $\mathcal{D}$ is the family

$$b_V = \mathbb{E}_{x \sim \mathcal{D}}[\chi_V(x)], \quad V \subseteq [n]$$

— the dataset Fourier coefficients of the constant function 1. Note $b_\emptyset = 1$ and $|b_V| \leq 1$. For $\theta \in (0, 1]$ — we never take θ larger — write $B_\theta = \{V : |b_V| \geq \theta\}$ for the **$\boldsymbol{\theta}$-heavy bias set**; since $b_\emptyset = 1$, always $\emptyset \in B_\theta$, so $|B_\theta| \geq 1$.

Lemma 2.5 *(Convolution Identity)*: For any $f : \{-1, 1\}^n \to \mathbb{R}$ and any $S \subseteq [n]$,

$$\hat{f}_{\mathcal{D}}(S) = \sum_{V \subseteq [n]} b_V \cdot \hat{f}(S \Delta V),$$

where Δ denotes symmetric difference.

Proof: Expand f in its global Fourier expansion inside the dataset expectation:

$$\hat{f}_{\mathcal{D}}(S) = \sum_T \hat{f}(T) \cdot \mathbb{E}_{x \sim \mathcal{D}}[\chi_T(x)\chi_S(x)] = \sum_T \hat{f}(T) \cdot b_{T \Delta S},$$

using $\chi_T \chi_S = \chi_{T \Delta S}$; substitute $V = T \Delta S$. ∎

The bias spectrum is exactly the aliasing structure of the subcube example: there $b_V = \mathbb{1}[V \subseteq K]$, and the convolution smears the single global coefficient $\hat{f}(\emptyset) = 1$ onto all $2^{|K|}$ sets $S \subseteq K$. This coefficient aliasing is the same mechanism that subsampling induces on the Walsh-

Hadamard spectrum, exploited constructively — with a *designed*, invertible subsampling pattern — in sparse-transform algorithms [30, 34]; the difference here is that the dataset, and hence its aliasing pattern, is given rather than chosen. The blindness of Lemma 2.2 is the statement that for a generic dataset the bias spectrum is an exponentially wide field of $\sim \pm 1/\sqrt{|\mathcal{D}|}$ noise, which buries any bucket-level signal. Conversely, whenever the bias spectrum is *dominated by few heavy terms*, the dataset spectrum of f is a controlled deformation of its global spectrum — which the following theorem exploits.

Theorem 2.3 (Dataset Goldreich-Levin, model-query access): Let $f : \{-1,1\}^n \to [-1,1]$ with $\|\hat{f}\|_1 \le A$ be given by QUERY access, and let $\mathcal{D}$ be given by SAMP access. Let $0 < \tau \le 1$, $\delta \in (0,1)$, fix any $\theta \le \tau/(4A)$, and suppose the θ-heavy bias set B_θ of $\mathcal{D}$ is given explicitly (e.g., via Theorem 2.2 applied to $f \equiv 1$, or from prior structural knowledge). Then there is an algorithm running in time $\mathrm{poly}(n, |B_\theta|, 1/\tau, \log(1/\delta))$, using $\mathrm{poly}(n, |B_\theta|, 1/\tau) \cdot \log(1/\delta)$ queries to f and $O(\log(|B_\theta| / (\tau\delta))/\tau^2)$ samples from $\mathcal{D}$, that with probability at least $1 - \delta$ outputs a list L satisfying:
1. (Completeness) $|\hat{f}_{\mathcal{D}}(S)| \ge \tau \Longrightarrow S \in L$;
2. (Soundness) $S \in L \Longrightarrow |\hat{f}_{\mathcal{D}}(S)| \ge \tau/2$;
3. $|L| \le \frac{64\,|B_\theta|^3}{9\tau^2}$.

The algorithm:
1. Run classical GL (Theorem 2.1, in its $[-1,1]$-valued form) on f with threshold $\tau' = \frac{3\tau}{4\,|B_\theta|}$ and confidence $1 - \delta/2$, obtaining a list F of global heavy sets, $|F| \le 4/\tau'^2$.
2. Form the candidate list $C = \{T \Delta V : T \in F, V \in B_\theta\}$, so $|C| \le |F| \cdot |B_\theta|$.
3. Draw $m = O(\log(|C| / \delta)/\tau^2)$ samples from $\mathcal{D}$ and, for each $S \in C$, compute the empirical coefficient $\tilde{f}_{\mathcal{D}}(S) = \frac{1}{m} \sum_{j=1}^m f(x^{(j)}) \chi_S(x^{(j)})$. Output $L = \left\{S \in C : |\tilde{f}_{\mathcal{D}}(S)| \ge 3\tau/4\right\}$.

Proof: **Decomposition.** By Lemma 2.5, for every S,
$$\hat{f}_{\mathcal{D}}(S) = \sum_{V \in B_\theta} b_V \hat{f}(S \Delta V) + R_S, \quad \text{where } |R_S| \le \sum_{V \notin B_\theta} |b_V|\ |\hat{f}(S \Delta V)| \le \theta \cdot \|\hat{f}\|_1 \le \theta A \le \frac{\tau}{4},$$
using $|b_V| < \theta$ off the heavy set and re-indexing the tail sum over $T = S \Delta V$.

Completeness. Suppose $|\hat{f}_{\mathcal{D}}(S)| \ge \tau$. Then
$$\sum_{V \in B_\theta} |b_V|\ |\hat{f}(S \Delta V)| \ge |\hat{f}_{\mathcal{D}}(S)| - |R_S| \ge \tau - \frac{\tau}{4} = \frac{3\tau}{4},$$
and since $|b_V| \le 1$, some $V^* \in B_\theta$ has $|\hat{f}(S \Delta V^*)| \ge \frac{3\tau}{4\,|B_\theta|} = \tau'$. By the completeness of classical GL (except with probability $\delta/2$), $S \Delta V^* \in F$, hence $S = (S \Delta V^*) \Delta V^* \in C$. By Hoeffding and a union bound over C (except with probability $\delta/2$), every empirical coefficient satisfies $|\tilde{f}_{\mathcal{D}}(S) - \hat{f}_{\mathcal{D}}(S)| \le \tau/8$; in particular $|\tilde{f}_{\mathcal{D}}(S)| \ge \tau - \tau/8 > 3\tau/4$, so $S \in L$.

Soundness. If $S \in L$ then $|\hat{f}_{\mathcal{D}}(S)| \ge 3\tau/4 - \tau/8 = 5\tau/8 \ge \tau/2$.

List size and complexity. $|L| \le |C| \le |F|\ |B_\theta| \le \frac{4}{\tau'^2}\ |B_\theta| = \frac{64\,|B_\theta|^3}{9\tau^2}$. Classical GL at threshold τ' costs $\mathrm{poly}(n, 1/\tau') = \mathrm{poly}(n, |B_\theta|, 1/\tau)$ queries; step 3 costs m samples and $m \cdot |C|$ arithmetic operations. ∎

Note that the algorithm uses only the *set* B_θ, never the values b_V; and the theorem is meaningful precisely in the distribution-shift regime where $\hat{f}_{\mathcal{D}}(S)$ and $\hat{f}(S)$ differ — the shift is routed through the dataset's heavy biases. Running the completeness argument purely combinatorially (Parseval bounds the number of τ'-heavy global coefficients by $1/\tau'^2$) also yields an unconditional structural bound:

Corollary 2.1 *(List-Size Bound under Sparse Bias Spectrum)*: If $f : \{-1,1\}^n \to [-1,1]$ with $\left\|\hat{f}\right\|_1 \le A$, and $\theta \le \tau/(4A)$, then

$$|\left\{S : |\hat{f}_{\mathcal{D}}(S)| \ge \tau\right\}| \le \frac{16\,|B_\theta|^3}{9\tau^2}.$$

Representative Datasets

At the opposite extreme, if the dataset's spectrum uniformly tracks the global one — the "no distribution shift" regime — dataset GL degenerates gracefully to classical GL plus re-scoring.

Definition 2.6 *(ϵ-Representative Dataset)*: $\mathcal{D}$ is **ϵ-representative** for f if $|\hat{f}_{\mathcal{D}}(S) - \hat{f}(S)| \le \epsilon$ for all $S \subseteq [n]$.

Lemma 2.6: Let $f : \{-1,1\}^n \to [-1,1]$ and let $\mathcal{D}$ consist of m i.i.d. uniform samples. If $m \ge \frac{2}{\epsilon^2}\left(n \ln 2 + \ln \frac{2}{\delta}\right)$, then with probability at least $1-\delta$, $\mathcal{D}$ is ϵ-representative for f.

Proof: For fixed S, $\hat{f}_{\mathcal{D}}(S)$ averages m i.i.d. terms $f(x)\chi_S(x) \in [-1,1]$ with mean $\hat{f}(S)$; Hoeffding gives $\Pr\left[|\hat{f}_{\mathcal{D}}(S) - \hat{f}(S)| > \epsilon\right] \le 2e^{-m\epsilon^2/2}$. Union bound over all 2^n sets. (Here $\mathcal{D}$ is the sampled *multiset* and $\hat{f}_{\mathcal{D}}(S)$ its multiset average; distinctness of points plays no role in this lemma or the proposition below.) ∎

Proposition 2.1 *(Dataset GL for Representative Datasets)*: Suppose $\mathcal{D}$ is $(\tau/8)$-representative for $f : \{-1,1\}^n \to [-1,1]$, and we have QUERY access to f and SAMP access to $\mathcal{D}$. Then the search problem of Definition 2.1 is solvable in time $\mathrm{poly}(n, 1/\tau, \log(1/\delta))$ with $O(\log(1/(\tau\delta))/\tau^2)$ samples: run classical GL at threshold $7\tau/8$ to get F, estimate $\hat{f}_{\mathcal{D}}(S)$ for $S \in F$ to accuracy $\tau/8$ from samples, and keep $|\tilde{f}_{\mathcal{D}}(S)| \ge 3\tau/4$.

Proof: If $|\hat{f}_{\mathcal{D}}(S)| \ge \tau$ then representativeness gives $|\hat{f}(S)| \ge \tau - \tau/8 = 7\tau/8$, so $S \in F$ by GL completeness; its estimate satisfies $|\tilde{f}_{\mathcal{D}}(S)| \ge |\hat{f}_{\mathcal{D}}(S)| - \tau/8 \ge 7\tau/8 > 3\tau/4$, so S is kept. (The representativeness slack is needed only for the GL step; the estimation step compares directly against $\hat{f}_{\mathcal{D}}(S)$.) If S is kept then $|\hat{f}_{\mathcal{D}}(S)| \ge 3\tau/4 - \tau/8 = 5\tau/8 \ge \tau/2$. GL's Parseval bound gives $|F| \le 4/(7\tau/8)^2 \le 7/\tau^2$, so a union bound over F needs $O(\log(|F|\,/\delta)/\tau^2)$ samples. ∎

In this regime the heavy dataset coefficients essentially *are* the heavy global coefficients; the substantive content of Theorem 2.2 and Theorem 2.3 lies exactly where representativeness fails.

2.6 Application: Kushilevitz-Mansour over Datasets

Classical GL powers the Kushilevitz-Mansour learning algorithm [29] ([31], Theorems 3.37-3.38): any f with $\left\|\hat{f}\right\|_1 \le s$ — in particular any decision tree of size s — is learnable from queries in time $\mathrm{poly}(n, s, 1/\epsilon)$. The spectral-norm hypothesis is exactly the A of Theorem 2.3 — the same L^1-geometry exploited by agnostic decision-tree learning [20] — so the on-distribution analogue of the coefficient-collection step comes for free:

Corollary 2.2 *(Heavy On-Dataset Coefficients of Decision Trees)*: Let $f : \{-1,1\}^n \to \{-1,1\}$ be computable by a decision tree of size s (so $\left\|\hat{f}\right\|_1 \le s$), given by QUERY access, and let B_θ be the θ-heavy bias set of $\mathcal{D}$ for some $\theta \le \tau/(4s)$. Then all S with $|\hat{f}_{\mathcal{D}}(S)| \ge \tau$ can be listed in time $\mathrm{poly}(n, s, |B_\theta|, 1/\tau, \log(1/\delta))$.

We emphasize what Corollary 2.2 does *not* yet give: an $\epsilon_{\mathcal{D}}$-close sparse approximation to f. Classically, coefficient collection plus Parseval yields the learning guarantee; over a dataset, the closeness identity is constant-free only in *normalized* coefficients (Theorem 1.2), and a sparse g assembled from recovered raw coefficients has dataset coefficients that differ from its design coefficients (aliasing again), so bounding $\mathbb{E}_{x\sim\mathcal{D}}[(f-g)^2]$ requires genuinely new arguments. TODO: On-dataset approximation from recovered coefficients. One route: $\mathbb{E}_{\mathcal{D}}[(f-g)^2] = \sum_V b_V \cdot \widehat{(f-g)^2}(V)$ by Lemma 2.5 applied to $(f-g)^2$ at $S = \emptyset$, splitting the sum over B_θ and its complement; the tail is controlled by $\theta \cdot \left\|\widehat{(f-g)^2}\right\|_1 \le \theta\left(\left\|\hat{f}\right\|_1 + \|\hat{g}\|_1\right)^2$, but the heavy-bias terms need per-V control. Compare with the frame-truncation route of Theorem 1.3.

2.7 Discussion and Open Directions

The most closely related work is the Active Fourier Auditor [2], which likewise estimates spectral properties of a model over a data distribution; it orthonormalizes the parity basis to that distribution (following [25]) and reports a few scalar functionals (robustness, individual/group fairness), whereas our dataset Goldreich-Levin keeps the fixed characters over the empirical measure and recovers the heavy coefficients themselves. See Section 3 for the fuller landscape (non-uniform Fourier, the samples-versus-queries barrier, subsampling aliasing, and Fourier auditing of trained models).

TODO:

- Non-uniform (weighted) datasets: the identities above should extend with $p(x)$ in place of $1/|\mathcal{D}|$; the reproducing-kernel steps are unaffected, the mass identity picks up $\|p\|_2^2$-type factors.
- General product alphabets and orthonormal bases (Hermite, tokens): the pair-form and blindness lemmas need only the reproducing kernel; the convolution identity needs the group structure ($\chi_S\overline{\chi_T} = \chi_{S-T}$), so state it over $\mathbb{Z}_q^n$ or general finite abelian groups. For token alphabets, CSAMP is literally n-gram retrieval, and Theorem 2.2 applies as stated.
- Choosing the split order: the profile R_k depends on the coordinate order; find the order minimizing $\max_k R_k$ (a combinatorial problem on the dataset's context statistics — related to building a good trie).
- CSAMP from a flat subsample only: estimating Ψ without a corpus index requires context collisions *within* the subsample — a birthday/U-statistic analysis; quantify the subsample size needed as a function of the fiber profile.
- Characterize which real datasets (images, token corpora) have small context profiles c_k along some order, and how N behaves empirically; relatedly, when is the heavy-bias set B_θ small and low-degree?
- Lower bound: turn Lemma 2.2 into a formal sample-complexity lower bound for Definition 2.1 in the SAMP-only model (two datasets, same visible statistics, different heavy sets), and extend it to CSAMP with all fibers singletons.

3 Related Work

Our starting point — the Fourier analysis of functions on $\{-1,1\}^n$ — is classical [31]. What changes here is the *measure*: we replace the uniform distribution on the cube by the uniform distribution on a finite dataset $\mathcal{D}$, *keep* the standard parity characters χ_S, and ask what of the Fourier toolkit survives. This single choice — fixed basis, empirical measure — is what forces the non-orthogonality defect $C_{\mathcal{D}} = 2^n / |\mathcal{D}|$, the bias-spectrum convolution, and the samples-only "blindness" barrier. We organize prior work around it.

Fourier analysis off the uniform measure. The best-understood non-uniform setting is the p-biased / product-measure analysis ([31], Chapter 8; for learning, [14]), where the basis is rebuilt to remain orthonormal, so Parseval still holds. A recent line pushes this to genuinely non-product distributions by *orthonormalizing the basis to the distribution*: the discrete Fourier expansion of [25] for correlated variables, its use for learning DNF under a fixed distribution [24], and, in uncertainty quantification, the arbitrary polynomial chaos expansion built from the empirical moments of a raw dataset [32]. All of these *re-orthonormalize* and thereby avoid any $C_{\mathcal{D}}$-type defect; we instead retain the fixed characters and confront the non-orthogonality directly, which is what makes the dataset spectrum a *convolution* of the global spectrum rather than a change of basis.

Heavy-coefficient recovery and access models. The engine we adapt is the Goldreich–Levin algorithm [18] and its use in the Kushilevitz–Mansour learner [29], extended to general finite abelian groups by [3]; the agnostic-learning refinements of [20] exploit exactly the bounded spectral-norm ($\|\hat{f}\|_1 \leq A$) geometry we assume. All of these operate under the *uniform* measure with query access; our results are the empirical-measure incarnation, and the same $\|\hat{f}\|_1$ hypothesis is what routes the dataset shift through the bias spectrum.

Why samples alone are insufficient. That correlational, sample-only access cannot isolate a heavy parity is the moral of the statistical-query and noisy-parity barriers [6, 7, 28]. The same phenomenon reappears in verification: PAC-verification and the "interactive Goldreich–Levin" of [19] and [22] let a sample-only verifier recover heavy coefficients *only* with the help of a query-capable prover. Our blindness lemma is a sharp, dataset-specific incarnation of this barrier (a birthday-horizon on collision-free projections). Where those works restore power through an interactive prover, we restore it either through *conditional (subcube) sampling* over the empirical dataset — the oracle studied for distribution testing and learning [9–11], here exposing the context repetition that real corpora exhibit — or, when the model is queryable, through the dataset's bias spectrum.

Aliasing as subsampling. The convolution identity $\hat{f}_{\mathcal{D}}(S) = \sum_V b_V \hat{f}(S \Delta V)$ is, mechanically, the aliasing that subsampling induces on the Walsh–Hadamard spectrum, exploited constructively in sparse-transform algorithms [30, 34] and in compressed-sensing recovery of sparse set functions [4, 36], including in explicitly non-orthogonal bases [37]. The distinction is that those methods *design* an invertible subsampling pattern with full query access, whereas our dataset is *given*: its induced aliasing (the bias spectrum) is not ours to choose, which is precisely why samples alone are blind and why the bias spectrum must be supplied.

Auditing and interpreting trained models via Fourier. Closest in motivation is the Active Fourier Auditor [2], which estimates robustness and (individual/group) fairness of a model from queries and samples, building on the fairness-verification line of [15–17] and the query-based auditing of [38]. Crucially, that work *also* leaves the uniform measure — but by Gram–Schmidt-orthonormalizing the parity basis to the data distribution (the expansion of [25]) and estimating a handful of scalar functionals of the spectrum; we keep the fixed characters over the empirical measure and target the spectrum itself. A parallel, independent thread recovers the *sparse Walsh–Hadamard spectrum of a trained model* for explanation [21, 27], and — nearest to our machinery — [33] searches an *in-distribution* activation support for non-redundant heavy coefficients via an explicit "Goldreich–Levin with problem-specific constraints"; dataset Goldreich–Levin is the missing recovery primitive underneath these applications.

Sensitivity, learnability, and testing. Our per-coordinate sensitivity $\mathrm{Sens}^i_{\mathcal{D}}$ is the Fourier form of the total-effect Sobol index of variance-based global sensitivity analysis [35], whose subtlety under *dependent* (non-product) inputs [26] is the classical shadow of the $C_{\mathcal{D}}$ corrections; the same variance-based influence, estimated over a dataset, underlies feature-attribution methods such as [12]. The question of whether a model class can fit a given dataset is distribution-specific learnability: learnability under a *fixed* measure is governed by its metric entropy [5], and, spectrally, by how the target's weight aligns with the data-dependent spectrum [1, 8] — an information-theoretic cousin being dataset difficulty relative to a model family [13]. Finally, our in-distribution testing (which zeroes points outside $\mathcal{D}$) is close in spirit to distribution-free property testing [23], measuring distance over the data distribution rather than the uniform one.

4 Conclusion and Future Directions

Taking expectations over a dataset while keeping the characters of the ambient product space yields a usable, sample-estimable Fourier calculus — provided one respects the density constant $C_{\mathcal{D}}$ and the aliasing it encodes. The dictionary (lift, inversion, mass identity, closeness) is elementary; the payoff is that classical spectral algorithms can be transplanted honestly: low-degree learning survives with a scaled reconstruction, and Goldreich-Levin survives exactly when the dataset's bias spectrum is dominated by few heavy terms — with a matching impossibility (blindness) for sample-only access.

Beyond the open problems listed at the end of the Goldreich-Levin section, some directions we find promising:

TODO:

- *Degree monotonicity.* Is there a monotone relationship between the degree (spectral concentration) of f over the dataset and over the full space? A clean statement here would let one read off, from dataset quantities alone, whether a given model class can learn f — connect to distribution-specific learnability under a fixed measure [5], to spectral notions of learnability via data-dependent eigenbases [1, 8], and to dataset difficulty relative to a model family [13].
- *The dataset as its own function.* What happens when $f = \mathcal{D}$ (the indicator itself)? The bias spectrum is then the whole story; relate its decay to standard notions of dataset complexity.
- *PAC learning and entropy.* Relate normalized weight profiles $\overline{W}^k_{\mathcal{D}}$ to PAC-learnability on-distribution, and to entropy-type quantities (an analogue of Hartley entropy for datasets).

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Bibliography

[1] Emmanuel Abbe, Enric Boix-Adserà, and Theodor Misiakiewicz. 2023. SGD learning on neural networks: Leap complexity and saddle-to-saddle dynamics. In *Proceedings of the 36th Annual Conference on Learning Theory (COLT)*, 2023.

[2] Ayoub Ajarra, Bishwamittra Ghosh, and Debabrota Basu. 2025. Active Fourier auditor for estimating distributional properties of ML models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.

[3] Adi Akavia, Shafi Goldwasser, and Shmuel Safra. 2003. Proving hard-core predicates using list decoding. In *44th Annual IEEE Symposium on Foundations of Computer Science (FOCS)*, 2003. 146–157.

[4] Andisheh Amrollahi, Amir Zandieh, Michael Kapralov, and Andreas Krause. 2019. Efficiently learning Fourier sparse set functions. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[5] Gyora M. Benedek and Alon Itai. 1991. Learnability with respect to fixed distributions. *Theoretical Computer Science* 86, 2 (1991), 377–389.

[6] Avrim Blum, Merrick Furst, Jeffrey Jackson, Michael Kearns, Yishay Mansour, and Steven Rudich. 1994. Weakly learning DNF and characterizing statistical query learning using Fourier analysis. In *Proceedings of the 26th Annual ACM Symposium on Theory of Computing (STOC)*, 1994. 253–262.

[7] Avrim Blum, Adam Kalai, and Hal Wasserman. 2003. Noise-tolerant learning, the parity problem, and the statistical query model. *Journal of the ACM* 50, 4 (2003), 506–519.

[8] Abdulkadir Canatar, Blake Bordelon, and Cengiz Pehlevan. 2021. Spectral bias and task-model alignment explain generalization in kernel regression and infinitely wide neural networks. *Nature Communications* 12, 1 (2021), 2914.

[9] Clément L. Canonne, Dana Ron, and Rocco A. Servedio. 2015. Testing probability distributions using conditional samples. *SIAM Journal on Computing* 44, 3 (2015), 540–616.

[10] Sourav Chakraborty, Eldar Fischer, Yonatan Goldhirsh, and Arie Matsliah. 2013. On the power of conditional samples in distribution testing. In *Proceedings of the 4th Conference on Innovations in Theoretical Computer Science (ITCS)*, 2013. 561–580.

[11] Xi Chen, Rajesh Jayaram, Amit Levi, and Erik Waingarten. 2021. Learning and testing junta distributions with subcube conditioning. In *Proceedings of the 34th Annual Conference on Learning Theory (COLT) (PMLR)*, 2021. 1060–1113.

[12] Anupam Datta, Shayak Sen, and Yair Zick. 2016. Algorithmic transparency via quantitative input influence: Theory and experiments with learning systems. In *2016 IEEE Symposium on Security and Privacy (SP)*, 2016. 598–617.

[13] Kawin Ethayarajh, Yejin Choi, and Swabha Swayamdipta. 2022. Understanding dataset difficulty with V-usable information. In *Proceedings of the 39th International Conference on Machine Learning (ICML) (PMLR)*, 2022.

[14] Merrick L. Furst, Jeffrey C. Jackson, and Sean W. Smith. 1991. Improved learning of AC0 functions. In *Proceedings of the 4th Annual Workshop on Computational Learning Theory (COLT)*, 1991. 317–325.

[15] Bishwamittra Ghosh, Debabrota Basu, and Kuldeep S. Meel. 2021. Justicia: A stochastic SAT approach to formally verify fairness. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2021.

[16] Bishwamittra Ghosh, Debabrota Basu, and Kuldeep S. Meel. 2022. Algorithmic fairness verification with graphical models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2022.

[17] Bishwamittra Ghosh, Debabrota Basu, and Kuldeep S. Meel. 2023. "How biased are your features?": Computing fairness influence functions with global sensitivity analysis. In *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2023.

[18] Oded Goldreich and Leonid A. Levin. 1989. A hard-core predicate for all one-way functions. In *Proceedings of the 21st Annual ACM Symposium on Theory of Computing (STOC)*, 1989. 25–32.

[19] Shafi Goldwasser, Guy N. Rothblum, Jonathan Shafer, and Amir Yehudayoff. 2021. Interactive proofs for verifying machine learning. In *12th Innovations in Theoretical Computer Science Conference (ITCS)*, 2021.

[20] Parikshit Gopalan, Adam Tauman Kalai, and Adam R. Klivans. 2008. Agnostically learning decision trees. In *Proceedings of the 40th Annual ACM Symposium on Theory of Computing (STOC)*, 2008. 527–536.

[21] Ali Gorji, Andisheh Amrollahi, and Andreas Krause. 2025. Amortized SHAP values via sparse Fourier function approximation. In *International Conference on Learning Representations (ICLR)*, 2025.

[22] Tom Gur, Mohammad Mahdi Jahanara, Mohammad Mahdi Khodabandeh, Ninad Rajgopal, Bahar Salamatian, and Igor Shinkar. 2024. On the power of interactive proofs for learning. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC)*, 2024.

[23] Shirley Halevy and Eyal Kushilevitz. 2007. Distribution-free property-testing. *SIAM Journal on Computing* 37, 4 (2007), 1107–1138.

[24] Mohsen Heidari and Roni Khardon. 2025. Learning DNF through generalized Fourier representations. *arXiv preprint arXiv:2506.01075* (2025).

[25] Mohsen Heidari, Jithin K. Sreedharan, Gil I. Shamir, and Wojciech Szpankowski. 2021. Finding relevant information via a discrete Fourier expansion. In *Proceedings of the 38th International Conference on Machine Learning (ICML) (PMLR)*, 2021. 4181–4191.

[26] Giles Hooker. 2007. Generalized functional ANOVA diagnostics for high-dimensional functions of dependent variables. *Journal of Computational and Graphical Statistics* 16, 3 (2007), 709–732.

[27] Justin Singh Kang, Landon Butler, Abhineet Agarwal, Yigit Efe Erginbas, Ramtin Pedarsani, Kannan Ramchandran, and Bin Yu. 2025. SPEX: Scaling feature interaction explanations for LLMs. *arXiv preprint arXiv:2502.13870* (2025).

[28] Michael Kearns. 1998. Efficient noise-tolerant learning from statistical queries. *Journal of the ACM* 45, 6 (1998), 983–1006.

[29] Eyal Kushilevitz and Yishay Mansour. 1993. Learning decision trees using the Fourier spectrum. *SIAM Journal on Computing* 22, 6 (1993), 1331–1348.

[30] Xiao Li, Joseph K. Bradley, Sameer Pawar, and Kannan Ramchandran. 2015. SPRIGHT: A fast and robust framework for sparse Walsh–Hadamard transform. *arXiv preprint arXiv:1508.06336* (2015).

[31] Ryan O'Donnell. 2021. Analysis of boolean functions. *arXiv preprint arXiv:2105.10386* (2021).

[32] Sergey Oladyshkin and Wolfgang Nowak. 2012. Data-driven uncertainty quantification using the arbitrary polynomial chaos expansion. *Reliability Engineering & System Safety* 106, (2012), 179–190.

[33] Kyle Reing, Greg Ver Steeg, and Aram Galstyan. 2025. Making sense of distributed representations with activation spectroscopy. *arXiv preprint arXiv:2501.15435* (2025).

[34] Robin Scheibler, Saeid Haghighatshoar, and Martin Vetterli. 2015. A fast Hadamard transform for signals with sublinear sparsity in the transform domain. *IEEE Transactions on Information Theory* 61, 4 (2015), 2115–2132.

[35] Ilya M. Sobol. 2001. Global sensitivity indices for nonlinear mathematical models and their Monte Carlo estimates. *Mathematics and Computers in Simulation* 55, 1–3 (2001), 271–280.

[36] Peter Stobbe and Andreas Krause. 2012. Learning Fourier sparse set functions. In *Proceedings of the 15th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2012. 1125–1133.

[37] Chris Wendler, Andisheh Amrollahi, Bastian Seifert, Andreas Krause, and Markus Püschel. 2021. Learning set functions that are sparse in non-orthogonal Fourier bases. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2021.

[38] Tom Yan and Chicheng Zhang. 2022. Active fairness auditing. In *Proceedings of the 39th International Conference on Machine Learning (ICML) (PMLR)*, 2022.

APPENDIX A Numerical Verification

Every identity, counterexample, and theorem constant stated in this paper has been verified numerically on random and structured instances, exact to 10^{-9}: the mass and closeness identities, the Parseval-frame identities, the pair-form and blindness lemmas, the convolution identity and its heavy-bias tail bound, all five properties of the conditional bucket weight Ψ together with the unbiasedness of its estimator, the subcube and half-cube aliasing examples, the dataset sensitivity bounds, the corrected learning bound, and the candidate-generation constants of both Goldreich-Levin theorems. The self-contained script (Python, standard library only) lives at `verification/verify_identities.py` in the repository; run it with `python3 verification/verify_identities.py`.

TODO: Deferred proofs and the general finite-abelian-group statements, as they accumulate.